Implementation and Evaluation of an Automated Pairs Trading Algorithm

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***Abstract*—Today automated trading systems are a considerable force in financial markets worldwide. These computerized trading systems are made up of algorithms which execute a strategy often so fast or complex that they would be impossible for a human trader. One such strategy, pairs trading, takes advantage of a computer to detect and monitor correlated stock pairs. This paper will review an implementation and evaluation of an automated pairs trading algorithm.**

I. Introduction

In the past decade automated trading systems have grown from a research topic to a considerable force in global financial markets. These computerized trading systems are built around algorithms that utilize arbitrage, statistical arbitrage, trend following and/or mean reversion to execute a strategy often outside the scope of what would be possible for a human trader. Modern automated trading systems can be broken into three main components: The data stream, strategy unit and execution interface [1]. As input to the algorithm, the data stream is most commonly a feed of live and/or historical stock ticker quotes, however there have also been efforts to utilize outside data sources(i.e. social networks and news feeds) to predict market changes [2]. The strategy unit is the subsystem which does the predicting. Often using large amounts of data, the strategy unit makes trade decisions according to the specification laid out in the algorithm. The flexibility of digital systems allows these algorithms to be structured to utilize complex neural networks, or output decisions on the order of microseconds , as the particular strategy demands [3]. The execution interface is the final layer which allows the decisions made to be received by a financial institution. FIX (Financial Information eXchange) is a widely used protocol for both buy and sell side transactions, and is lightweight enough to be a reasonable choice for high frequency strategies [1] . This study will focus on the strategy unit, describing the implementation and performance of an automated pairs trading scheme. Pairs trading is a form of statistical arbitrage that is profitable in both bull and bear markets. The strategy begins with identification of stock pairs that are tightly correlated. These stocks are monitored for a weakening in the correlation, which creates a potentially profitable opportunity under the assumption that the ratio between the stocks is mean reverting. When such an opportunity arises, a bet is placed on their eventual convergence by longing the [proportionally] higher stock and short selling the lower stock [4] This strategy was chosen as it is market neutral, low risk, and the tasks of detecting correlations and monitoring stock pairs is ideal to be offloaded to a computer.

II. Pairs Detection

Detecting and monitoring the right pair(s) of stocks is critical to the overall performance of this strategy. For pairs detection, a year of EoD ticker data for all symbols belonging to NASDAQ, NYSE (including AMEX), and COMEX was downloaded from the ichart.finance.yahoo server using a simple python script. A desirable pair will exhibit proportionately similar motions in the same direction most of the time, but will occasionally exhibit independent motion in order to allow for profitable opportunity. After much fine tuning, the following formula was found to be able to detect if two stocks were a quality pair:

For potential pair of stocks whose end of day percent of YTD max is S_1 and S_2, and whose respective daily percentage change is S_1', S_2' we create the signal

$$x[n] = \sum_{m=1}^{365} \log\left(S_1'[m]S_2'[k]\right) \log\left(\frac{1}{|S_1[m] - S_2[k]|}\right) \quad (1)$$

where

$$k = m - n + \lceil N/2 \rceil$$

The unitless signal recorded in x[n] provides a quantitative measure of pair quality as a function of lags in the second signal. In this experiment N = 5 , to allow for up to 2 days of lag in either direction. By brute force, x[n] was found for the over 31 million possible stock pairs. Split

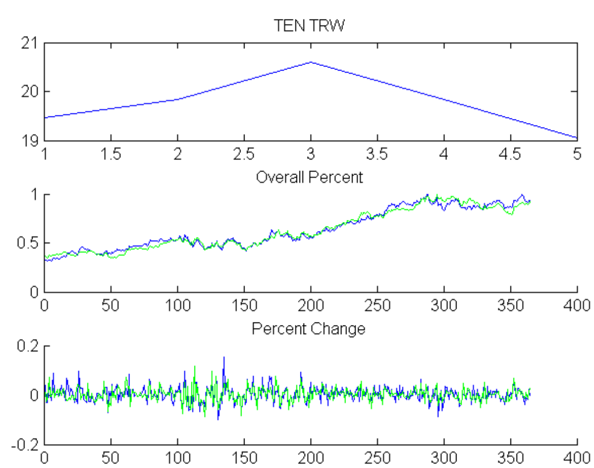

Fig. 1. From top to bottom: x[n], S and S' signals for pair Tennco Inc. and TRW Automotive Holdings Corp.

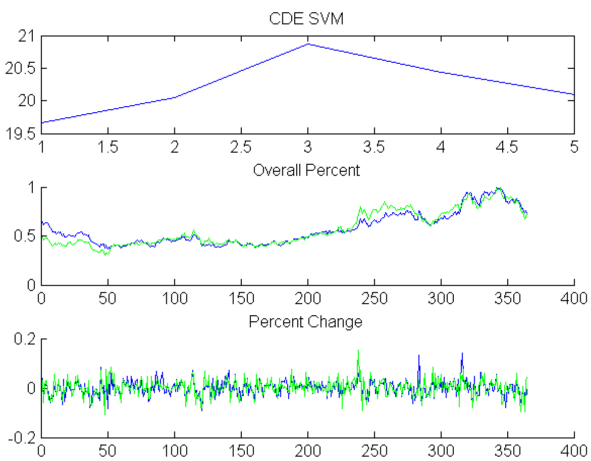

Fig. 2. From top to bottom: x[n], S and S' signals for pair Coeur d'Alene Mines Corporation and Silvercorp Metals Inc.

to run in parallel across four 2.4GHz processor cores, this operation took approximately 20 hours. The 100 signals with highest peak value were stored as candidates for pairs in the strategy unit. Figures 1 and 2 show examples of the pairs that this formula selects for. As would be expected, pairs arise from similar industry sectors. TEN and TRW are both automotive systems suppliers, while CDE and SVM are both primarily silver mining corporations.

III. Strategy

The *WealthLab Pro* strategy development and backtesting suite (offered by Fidelity) was used to implement the pairs trading scheme. WealthLab Pro offers a C# programming interface with a full API to facilitate the design and simulation of custom algorithmic trading systems. For simplicity, the strategy will be discussed in terms of a single stock pair, however the procedure could easily be repeated for a multi-pair trading strategy.

First the a moving average finite impulse response filter is run on the ratio of the the two unadjusted stock prices:

For unadjusted pair of EoD stock prices s_1 and s_2

$$r_{sma}[n] = \sum_{m=1}^{MAP} \frac{1}{MAP} \frac{s_1[n-m]}{s_2[n-m]} \tag{2}$$

where moving average period MAP is kept as an adjustable parameter. This is repeated for Δr such that

$$\Delta r[n] = \frac{s_1[n]}{s_2[n]} - r_{sma}[n] \tag{3}$$

and

$$\Delta r_{sma}[n] = \sum_{m=1}^{MAP} \frac{1}{MAP} \Delta r[n-m] \tag{4}$$

The signal is finally normalized by standard score

$$\Delta z[n] = \frac{\Delta r[n] - \Delta r_{sma}[n]}{\sigma_{\Delta r}} \tag{5}$$

where $\sigma_{\Delta r}$ is the standard deviation taken over the moving average period [5]. For The CDE SVM pair, this produced the signal shown in in Fig. 3. Using this signal as the core metric, the following strategy was employed:

for all *day* **do**
  if *positionsHeld* AND $z[n]$ crosses 0 **then**
    SELL all long positions
    COVER all short positions
  end if
  if $z[n] > highThresh$ **then**
    BUY stock2
    SHORT stock1
  else if $z[n] < lowThresh$ **then**
    BUY stock2
    SHORT stock1
  end if
end for

This is a simple aggressive algorithm that executes a hedged buy/sell order on the pair for each day z[n] exceeds either of the parametrized threshold values. It holds the positions until $z[n]$ returns to zero or changes sign. There there is a small risk that s_1 and s_2 enter a permanently diverging state causing orders to only be entered and never exited, however this was not encountered with any of the pairs selected.

IV. Backtesting Results

The to evaluate performance of the algorithm, trade execution was simulated using historical price data in the WealthLab Pro backtesting engine. The simulation was configured to execute each order for \$5000 in shares (making \$10,000 per pair trade), and assumed an unlimited amount of available capital. Commission rates were fixed at \$7.95 per trade to accurately model the cost of execution using a Fidelity account. Due to the nature of the algorithm, it was found that a span of 6 months to 1 year was needed to allow it to be effective. For all of the pairs tested, the trading scheme was found to be profitable, yet

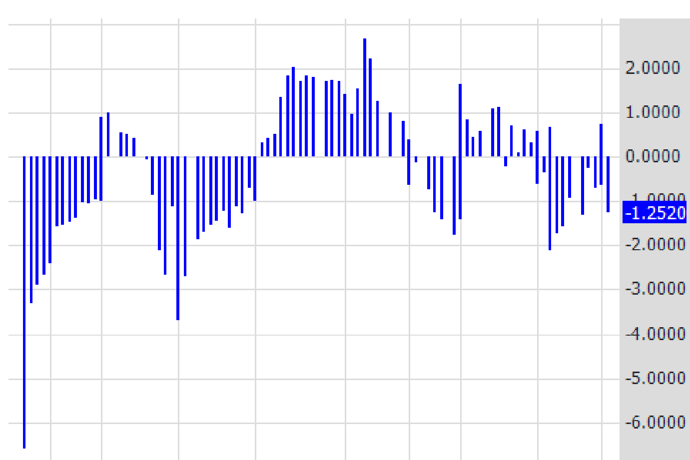

Fig. 3. $\Delta z[n]$ for CDE SVM pair

this was highly dependent on proper tuning of parameters MAP $highThresh$ and $lowThresh$. Fig. 4 shows backtest results for CDE SVM pair with $MAP = 50$ and $highThresh = -lowThresh = 1.34$. While there is an overall positive return for these parameter values, profits of this shape are not guaranteed for all ranges of the historical data set.

Backtest Performance Report | Range: 5/10/2010 to 5/9/2011 (253 Bars) |

	All Trades	Long Trades	Short Trades	Buy & Hold
Net Profit	$7,015.58	$49,540.23	($42,524.65)	$250.08
Profit per Bar	$1.92	$27.12	($23.28)	$0.99
Total Commission	($2,194.20)	($1,097.10)	($1,097.10)	($7.95)
Number of Trades	138	69	69	1
Average Profit	$50.84	$717.97	($616.30)	$250.08
Average Profit %	1.08%	14.34%	-12.18%	5.05%
Average Bars Held	26.48	26.48	26.48	252.00
Winning Trades	70	47	23	1
Win Rate	50.72%	68.12%	33.33%	100.00%
Gross Profit	$65,877.35	$56,239.12	$9,638.23	$250.08
Average Profit	$941.11	$1,196.58	$419.05	$250.08
Average Profit %	18.76%	23.91%	8.26%	5.05%
Average Bars Held	26.33	27.83	23.26	252.00
Max Consecutive Winne...	2	39	14	1
Losing Trades	68	22	46	0
Loss Rate	49.28%	31.88%	66.67%	0.00%
Gross Loss	($58,861.77)	($6,698.89)	($52,162.88)	$0.00
Average Loss	($865.61)	($304.50)	($1,133.98)	$0.00
Average Loss %	-17.12%	-6.09%	-22.40%	0.00%
Average Bars Held	26.63	23.59	28.09	0.00
Max Consecutive Losses	2	14	39	0
Maximum Drawdown	($12,609.62)	($21,938.06)	($55,925.33)	($1,243.62)
Maximum Drawdown D...	12/3/2010	1/24/2011	12/3/2010	7/19/2010
Profit Factor	1.12	8.40	0.18	Infinity
Recovery Factor	0.56	2.26	0.00	0.20
Payoff Ratio	1.10	3.92	0.37	0.00

Fig. 4. Backtest results for CDE SVM pair

V. Areas for Further Development

While this study has so far shown that this type of algorithm has the potential for profitability, there is still much room for improvement. One feature immediately recognizable from the simulated results (which also repeated itself in a number of other pairs tests), is that the losses from the short trades nearly eclipse gains made elsewhere. While in a single trade this would likely be accepted part of the nature of a hedged bet, the regularity by which it occurs in the simulation makes it a good candidate for further optimization. In this algorithm the hedge ratio was kept at 1:1; that is one quantity of long buy per one order of short sell. The upward overall market shift, which is suspected to have caused the disproportionate losses in the short positions, could have been estimated with another trend following algorithm and used to suggest a more appropriate hedge ratio. Taken to an extreme, the short orders could simply be eliminated, which would turn the algorithm into a heavily biased bull market strategy. This would operate on the theory that divergences are generally caused by just one stock temporarily failing to follow its partner in an upward movement. Conversely, a different approach to cut down on these losses would be to implement a 'half life' for positions held. The half life would be a period of time that the $\Delta z[n]$ is given to return to zero, after which the algorithm would exit the positions regardless of reconvergence. Although this also reduces overall risk, it is an unbiased addition which may cut into profits made by the long positions.

Furthermore, for practical reasons, the data used and strategy developed was all in terms of end of day pricing. With additional computational power, the data resolution could be increased to include intraday pricing. The trading strategy would be modified to take advantage of small pairs divergences within the span of a day. It would be very interesting to see a profitability comparison between long term and day trading implementations of this strategy.

As previously mentioned, scaling the algorithm to operate on multiple stock symbols is relatively trivial. Some development effort went into creating and backtesting such a strategy, however, in addition to scaled commission costs, it is much more difficult to analyse the behaviour in this case. Provided sufficient available capital, ultimately the most reliably profitable pairs trading scheme will diversify across a list of correlated pairs.

Perhaps the most damning shortcoming in the algorithm described is the fact that no set of parameter values (MAP, $highThresh$, $lowThresh$) was found to be capable of generating a consistent return on all subsets (of given length) of the historical data. This gives reason to doubt that a set of parameter values that are well tuned to have generated simulated profit in the previous 6 months, will indeed be profitable if applied for the following 6 months. To correct this, further investigation into each of the parameter values and their relationship to price data is needed. With more intelligent selection, or dynamic generation of parameter values, the pairs trading algorithm could become a much more reliable automated trading strategy.

VI. conclusion

This study has demonstrated mathematical tools capable of detecting if a pair of stocks are properly correlated to be useful in a pairs trading strategy. It has reviewed an implementation using a commercially available API from a major investment firm. While it was found that the algorithm is capable of generating positive profits in many circumstances, these profits have not been shown to be reliable. Modifications including diversification, a hedge ratio, changes to the strategy, and parameter tuning may improve the algorithm's reliability and profitability in future research.

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